

No. 92 plant: $\dot{y} = \theta_1 + \theta_2 y^3 + \theta_3 u$, $\theta_3 \geq \theta_0 > 0$, θ_0 is known

Refer: $\dot{y}_m = -2y_m + 2g$

Step 1. $\varepsilon = y_m - y$, $\dot{\varepsilon} = \dot{y}_m - \dot{y} = (-2y_m + 2g) - (\theta_1 + \theta_2 y^3 + \theta_3 u)$

$$\Rightarrow u = \frac{1}{\hat{\theta}_3} [-\hat{\theta}_1 - \hat{\theta}_2 y^3 + 2g - 2y_m + k\varepsilon] \quad (\hat{\theta}_i \text{ are estimates, } k > 0)$$

$$\tilde{\theta}_i = \theta_i - \hat{\theta}_i, \quad i=1,2,3 \Rightarrow \dot{\varepsilon} = -\frac{\theta_3}{\tilde{\theta}_3} k\varepsilon + \tilde{\theta}_1 + \tilde{\theta}_2 y^3 - \tilde{\theta}_3 u.$$

Step 2. Stability Analysis.

$$V(\varepsilon, \tilde{\theta}_1, \tilde{\theta}_2, \tilde{\theta}_3) = \frac{1}{2} \varepsilon^2 + \frac{1}{2\gamma_1} \tilde{\theta}_1^2 + \frac{1}{2\gamma_2} \tilde{\theta}_2^2 + \frac{1}{2\gamma_3} \tilde{\theta}_3^2,$$

$$\dot{V} = \varepsilon \dot{\varepsilon} + \frac{1}{\gamma_1} \tilde{\theta}_1 \dot{\tilde{\theta}}_1 + \frac{1}{\gamma_2} \tilde{\theta}_2 \dot{\tilde{\theta}}_2 + \frac{1}{\gamma_3} \tilde{\theta}_3 \dot{\tilde{\theta}}_3, \text{ since } \dot{\tilde{\theta}}_i = -\dot{\hat{\theta}}_i$$

$$\Rightarrow \dot{V} = -\frac{\theta_3}{\tilde{\theta}_3} k\varepsilon^2 + \tilde{\theta}_1 \left(\varepsilon - \frac{1}{\gamma_1} \dot{\hat{\theta}}_1 \right) + \tilde{\theta}_2 \left(\varepsilon y^3 - \frac{1}{\gamma_2} \dot{\hat{\theta}}_2 \right) + \tilde{\theta}_3 \left(-\varepsilon u - \frac{1}{\gamma_3} \dot{\hat{\theta}}_3 \right)$$

Step 3, choose $\dot{\hat{\theta}}_1 = \gamma_1 \varepsilon$, $\dot{\hat{\theta}}_2 = \gamma_2 \varepsilon y^2$, $\dot{\hat{\theta}}_3 = -\gamma_3 \varepsilon u$.

$$\hat{\theta}_3(t) \geq \theta_0 > 0 \Rightarrow \dot{\hat{\theta}}_3 = \begin{cases} -\gamma_3 \varepsilon u, & \text{if } (\hat{\theta}_3 > \theta_0) \text{ or } (\hat{\theta}_3 = \theta_0 \text{ and } -\varepsilon u \geq 0) \\ 0, & \text{otherwise.} \end{cases}$$

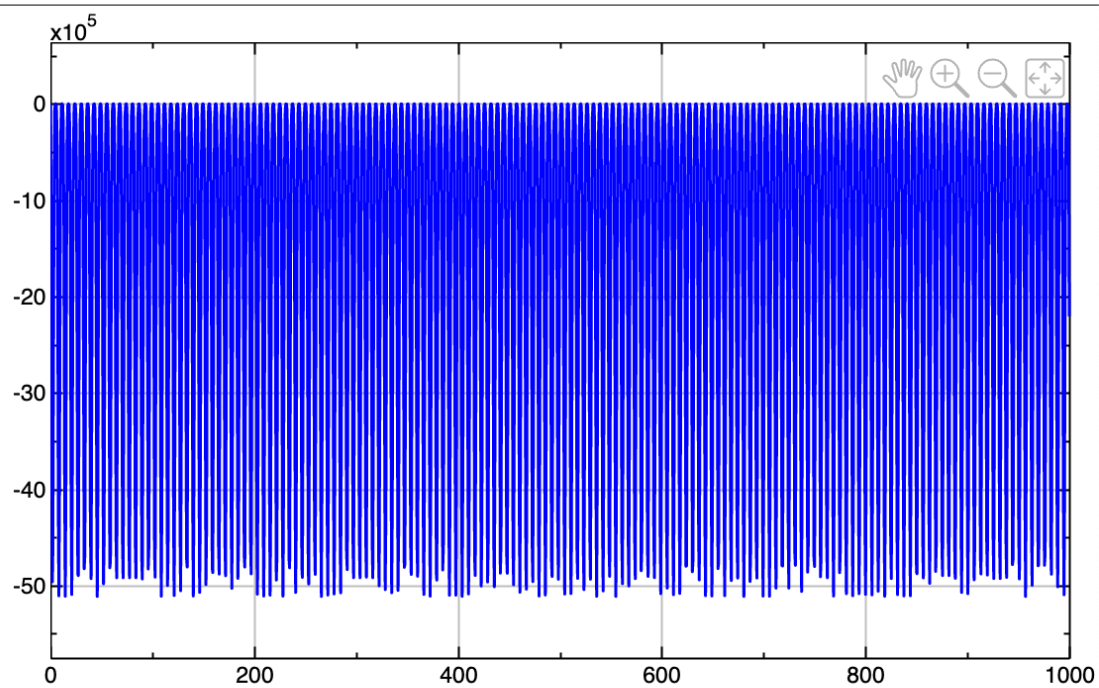
$$\Rightarrow \dot{V} = -\frac{\theta_0}{\hat{\theta}_3} k\varepsilon^2 \quad \text{it holds that } \dot{V} \leq -\frac{\theta_0}{\hat{\theta}_3} k\varepsilon^2 \leq 0.$$

Boundedness: Since $V(t) \geq 0$, $\dot{V}(t) \leq 0$, $V(t)$ is non-increasing and bounded

hence ε , $\tilde{\theta}_1$, $\tilde{\theta}_2$, $\tilde{\theta}_3$ are bounded $\rightarrow \hat{\theta}_i$ and y are bounded $\rightarrow u$ is also bounded.

Convergence of ε : $\dot{V} \leq -c\varepsilon^2$ with $c = \frac{\theta_0}{\hat{\theta}_3} k > 0$, $\rightarrow \varepsilon \in \mathcal{L}_2 \rightarrow \lim_{t \rightarrow \infty} \varepsilon(t) = 0$

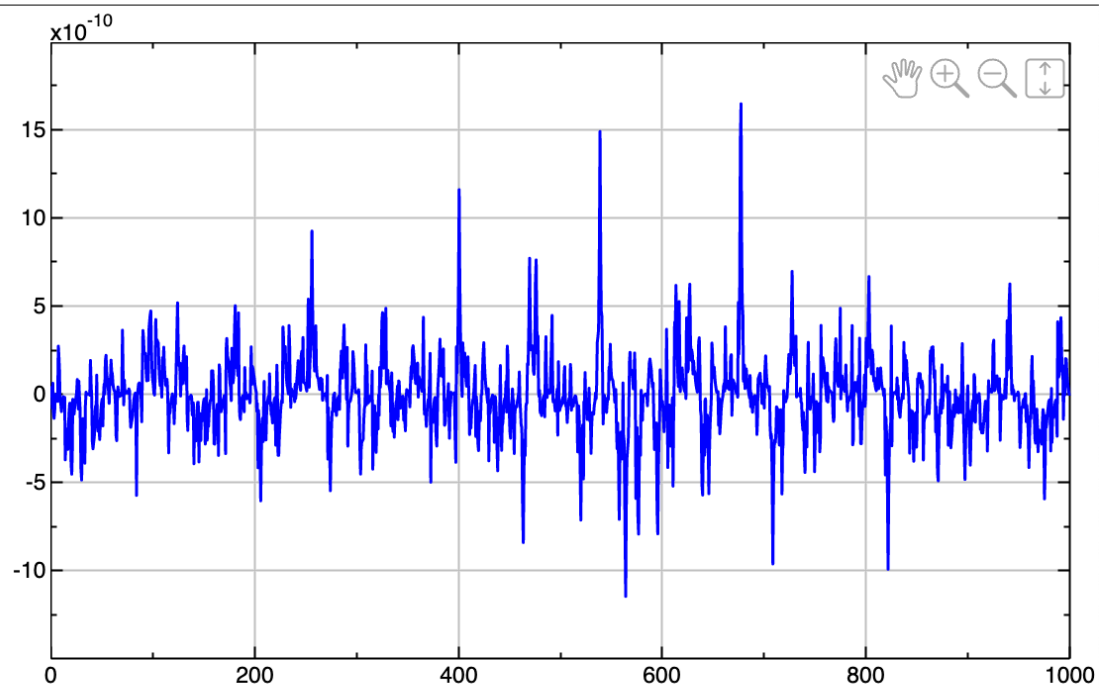
Adaptive Controller/u



就绪

基于采样 T=1000.0000

Adaptive Controller/epsilon



就绪

基于采样 T=1000.0000